Enhancement of Endoscopic Images and Video Using Advanced Deep Learning and Traditional Image Enhancement Techniques

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Abstract. Endoscopic images often struggle with issues like low contrast, noise, and lack of detail, making it harder for doctors to make accurate diagnoses. To help overcome these problems, our team developed a new image enhancement method using deep learning techniques. The system combines a Dual-Input U-Net with a Super-Resolution Convolutional Neural Network (SRCNN). The SRCNN takes a low-quality, low-resolution image and produces a clearer, higher-resolution version. Both this improved image and the original are then processed together by the Dual-Input U-Net for further refinement. After that, we apply several post-processing steps such as removing noise, adjusting brightness and contrast, fine-tuning lighting, and sharpening to make the images even clearer. We test how well this system works using metrics like Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM), which show our method outperforms traditional techniques. Overall, our results suggest that this approach can greatly improve endoscopic images, potentially helping doctors make more accurate diagnoses.

Keywords: Endoscopic Imaging, SRCNN, Dual-Input U-Net, Deep Learning, Image Enhancement, Medical Imaging, Post-Processing, SSIM, PSNR.

1 Introduction

Endoscopic imaging has really become an essential part of modern medical diagnostics, especially when it comes to checking out the gastrointestinal tract, lungs, and other internal organs. That said, even though it's widely used, traditional endoscopic images often leave something to be desired they can be blurry, noisy, or have poor contrast. These issues usually happen because of limitations in the camera hardware, not-so-great lighting conditions, or slight movements during the procedure, which can lead to blurry or distorted pictures [1, 2]. To fix this, people have turned to techniques like histogram equalization and adaptive filtering, but these often struggle to preserve all the delicate details and can sometimes make images look unnatural [11, 12]. Recently, advances in deep learning have opened up exciting new possibilities for improving

these images more intelligently and effectively [6]. Methods such as SRCNNs and U-Net models have shown promising results in making medical images clearer, sharper, and more detailed, which really helps doctors see better during exams [16, 18]. What's more, these models can be integrated into real-time systems, providing instant image enhancements during procedures. In this project, we introduce a new deep learning method that combines the strengths of SRCNNs with a Dual-Input U-Net architecture to boost endoscopic image quality. The SRCNN part is used to upscale and sharpen low-resolution images, while the Dual-Input U-Net refines the textures and structures by analyzing both the original and the enlarged images together [9, 11, 12]. To make sure that the model retains true-to-life details and doesn't lose important features, we developed a custom loss function that blends L1 loss, MSE, and an edge-aware component [16, 21]. After the deep learning processing, additional post-processing steps like sharpening, lighting correction, gamma adjustment, and noise reduction are applied to further enhance image clarity and usefulness for diagnosis [3,4,19,18]. We evaluated our approach using common image quality metrics like PSNR and SSIM which are widely used in medical image analysis [7, 8]. The results show that our combined method considerably improves image quality, emphasizing its strong potential for clinical use and real-time diagnostic systems [1, 10].

2 Literature Review

Using deep learning to analyze endoscopic images is quickly becoming an exciting way to assist with diagnosis, improve image quality, and help doctors make better decisions. Recently, techniques like convolutional neural networks (CNNs) have taken the lead, especially when it comes to picking out important features and classifying images [9, 11]. Popular models like ResNet [9] and DenseNet [11] are favorites because they make training smoother and more efficient. Adding attention mechanisms has also helped models zero in on the most relevant details, which has boosted their accuracy even more [16]. To handle the tricky problem of limited labeled data, methods like transfer learning (TL) and domain adaptation (DA) are being used successfully, making these tools way more practical in real clinical settings [14, 20]. While traditional image enhancement techniques still have a place, deep learning has brought in more advanced options, like generative adversarial networks (GANs) [19], which are really good at reducing noise and sharpening images. Recent progress in depth and pose estimation, often driven by self-supervised learning, is also tackling the issue of not having enough labeled depth data [23]. Alongside all these innovations, ensemble learning methods have shown they can make models more reliable, especially for classification tasks [15]. All these developments emphasize how quickly endoscopic image processing is moving forward and point to some really exciting directions for future research in the field.

Table 1. COMPARATIVE ANALYSIS OF LATEST RESEARCH STUDIES

Types of Models	Key Contribution	Strengths	Limitations
CNNs	Automatic feature extraction	Strong pattern recognition	Requires large datasets
ResNet	Addresses vanishing gradients	Enables deep networks	Risk of overfitting
Transfer Learning	Uses pre-trained models	Helps to minimize reliance on manually labeled data.	Domain similarity affects performance
GANs	Image enhancement & artifact removal	High-quality synthesis	Requires careful tuning
Self-Supervised Learning	Learns from unlabeled data	Reduces annotation dependency	Lower accuracy vs. supervised learning
Ensemble Learning	Combines multiple models	Improves robustness	Higher computational cost

3 Research Methodology

3.1 Dataset Description and Preprocessing Details

To test and validate our deep learning-based image enhancement method, we used five publicly available endoscopic datasets [3, 4, 17, 22, 25]. These datasets include images representing a wide range of gastrointestinal issues like polyps, ulcers, and inflammation, which helps ensure our model's performance is both reliable and relevant to real clinical scenarios. They also feature different imaging conditions such as poor lighting, motion blur, and variations in anatomy, which is essential for making sure our approach works well across different situations [2, 5, 19]. Plus, since these datasets are commonly used in recent research on endoscopic image enhancement and classification, it makes it easier to compare our results with the best existing methods [16, 20]. We've summarized key details such as resolution, dataset size, and types of annotations in Table 2, while Figure 1 displays sample images that emphasize the variety of pathologies and visual challenges faced in endoscopic imaging.

Table 2. SUMMARY OF ENDOSCOPIC IMAGE DATASETS

Name	Size
Kvasir SEG [22]	44.1 MB
Kvasir - Computer-Aided Gastrointestinal Disease Detection [22]	2.3 GB
GastroEndoNet [25]	407 MB
Endoscopic Real & Synthetic: Over and Underexposed Frames for Image Enhancement [3, 4]	192 MB
Endoscopy Artefact Detection (EAD) Dataset [17]	200 MB

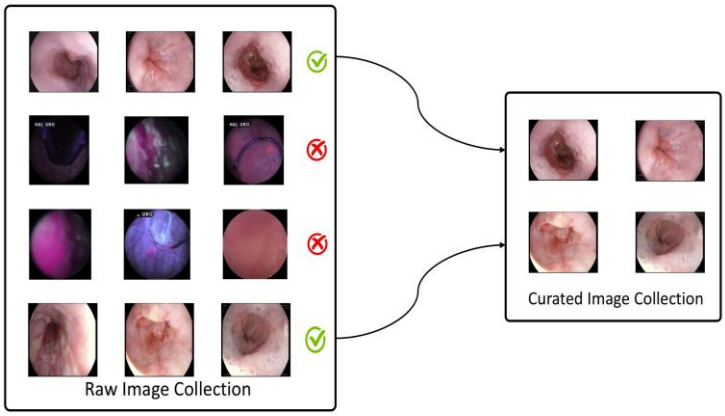

Fig. 1. Sample images from the datasets display a variety of gastrointestinal conditions along with associated artifacts.

All images are resized to $299 * 299$ Pixels and standard normalization techniques are applied during preprocessing to ensure consistency before feeding them into the model.

3.2 Deep Learning-Based Enhancement

The enhancement framework is built around a two-step deep learning setup. It starts with an SRCNN to boost the resolution, followed by a Dual-Input U-Net that fine-tunes the structure and texture details [9, 11]. The segregation between operations establishes a process for the system to improve low-quality endoscopic images through SRCNN because of its proven success in medical image resolution enhancement tasks [6, 12]. During Dual-Input U-Net processing, both the original image and the higher-resolution version are analyzed together to enhance anatomical details. This approach uses multi-scale features and combines information from both inputs to improve clarity [16]. The framework integrates multiple stages to maintain detailed image structures since it reduces typical distortions that conventional endoscopic imaging encounters [3, 4, 5].

SRCNN

An SRCNN model functions in the proposed framework to develop high-frequency details by processing initial low-resolution inputs [9]. As illustrated in Figure 2, the network comprises 3 Sequential convolutional layers, each serving a distinct function within the super-resolution process:

- Feature Extraction Layer: A $9 * 9$ Convolutional layer is used to extract low-level features from the input image, capturing essential patterns and structural cues.

- Non-linear Mapping Layer: This layer applies a $5 * 5$ Convolution to map the extracted features into a higher-dimensional feature space, enabling the network to learn complex representations.
- Reconstruction Layer: The reconstruction layer uses a second $5 * 5$ convolution to rebuild the high-resolution image from the processed feature maps, helping to recover fine textures and preserve spatial details.

This simple yet effective architecture enables the model to learn the end-to-end mapping between low and high resolution images directly, which has proven highly effective in various medical imaging contexts [6, 12].

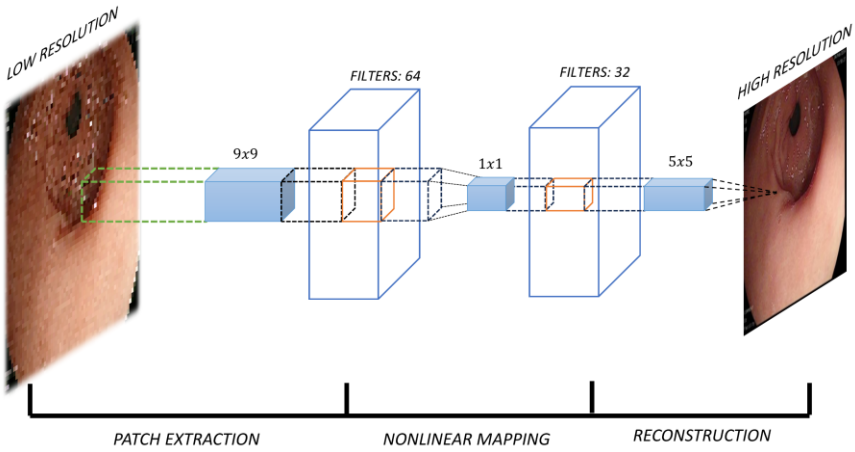

Fig. 2. The architecture of the SRCNN model is used for super-resolution enhancement.

Dual-Input U-Net

The intermediate output of the SRCNN is then mixed with the original low-resolution image and fed into a U-Net model for further enhancement of structural integrity [9, 16]. As illustrated in Figure 3, the U-Net architecture is organized into three primary components:

- **Encoder**: This segment captures global contextual features by applying a series of convolutional layers followed by max pooling operations, effectively reducing spatial dimensions while preserving essential information.
- **Bottleneck**: This part of the network focuses on extracting broad, high-level features that help it recognize and learn more complex patterns from the combined input.
- **Decoder:** The decoder combines upsampling methods with skip connections from the matching encoder layers to effectively recover detailed features. This approach ensures precise structural restoration and improves overall image quality.

This design combines the strengths of both SRCNN and U-Net, making sure that the images not only look sharper but also keep important details that matter for accurate clinical analysis [6, 12].

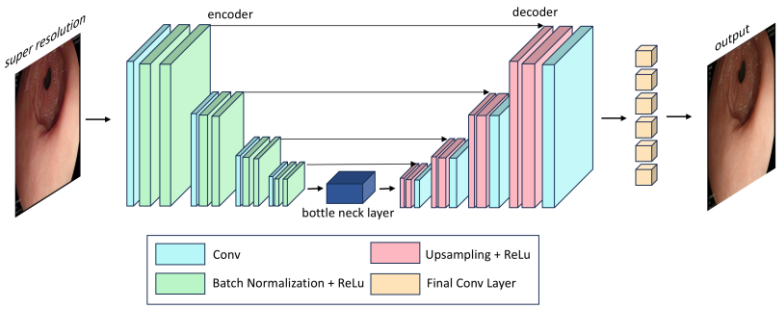

Fig. 3. Dual-Input U-Net architecture with skip connections and encoder-decoder structure.

To optimize training and ensure both perceptual quality and structural accuracy in the enhanced endoscopic images, a composite loss function is employed. This loss integrates multiple components, each contributing to a different aspect of image quality:

- L1 Loss (Mean Absolute Error): Encourages pixel-wise consistency between the predicted and ground truth images by minimizing absolute intensity differences.
- MSE Loss (Mean Squared Error): Focuses on reducing overall reconstruction errors by penalizing larger deviations more heavily, promoting smoother outputs.
- Edge Loss: The method relies on gradient differences calculated with a Sobel filter to emphasize edges, which helps maintain the clarity of fine anatomical boundaries and structural details [11, 16].

The total loss function is defined as:

$$L_{total} = \alpha L_1 + \beta L_{MSE} + \gamma L_{edge} \qquad (1)$$

where α, β, γ are weighting factors.

Post-Processing Techniques

To further enhance image clarity, contrast, and diagnostic quality, the following post-processing techniques are applied as depicted in Table 3:

Table 3. Overview of Post-Processing Methods and Their Purposes

Post-Processing Method	Purpose
Sharpening	Enhances fine details using a $3 * 3$ kernel
Illumination Enhancement	Adjusts contrast using CLAHE in LAB color space
Gamma Correction	Non-linear brightness adjustment via gamma transformation
Denoising	Reduces noise using Non-Local Means filtering

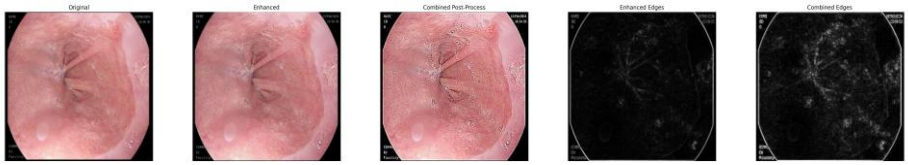

Fig. 4. Comparison of raw images and enhanced images after post-processing.

Enhancement Pipeline

The complete enhancement process is organized into a structured pipeline, as depicted in Figure 5. Each stage is designed to contribute to the progressive improvement of image quality and diagnostic usability:

1. **Preprocessing:** All dataset images are loaded and resized to a uniform resolution of $299 * 299$ Pixels. Normalization is then applied to ensure consistent input scaling across the pipeline.
2. **Super-Resolution:** The SRCNN model performs initial enhancement by reconstructing high-frequency image details and improving resolution.
3. **Structural Refinement:** A Dual-Input U-Net processes both the original low-resolution image and the SRCNN-enhanced output, refining structural textures and preserving anatomical integrity.
4. **Post-Processing:** A set of enhancement techniques is applied, including sharpening, illumination correction, and gamma adjustment. These steps further refine image clarity and visual quality.
5. **Evaluation:** Image quality assessment relies on PSNR alongside SSIM as standard performance metrics for measuring enhancement output effectiveness [6, 12].

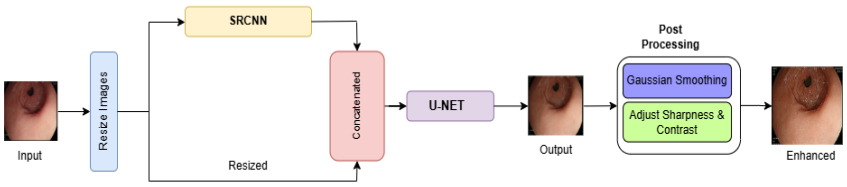

Fig. 5. Flowchart of the proposed deep learning based endoscopic image enhancement framework

The deep learning-based enhancement framework comprises three essential components: SRCNN for resolution enhancement [9], Dual-Input U-Net for structural refinement [16], and various post-processing methods that generate high-quality endoscopic images free from artifacts. These combined elements lead to noticeable improvements in image quality, including sharper detail, higher resolution, and clearer features. Tests

have shown important gains in both PSNR and SSIM, which confirms that this approach is effective [6, 12]. Overall, this method marks an important step forward in medical image enhancement, showing strong potential for use in real-time diagnostics and clinical workflows.

3.3 Comparative Novelty Analysis

Our proposed enhancement framework introduces an innovative approach by combining a Super-Resolution Convolutional Neural Network (SRCNN) with a Dual-Input U-Net to enhance both the resolution and structural accuracy of endoscopic images. While individually, SRCNN and U-Net architectures have been used in medical image enhancement, their sequential integration—first employing SRCNN to boost spatial resolution, followed by a dual-input U-Net for structural refinement—represents a new advancement in the field. Traditional enhancement methods usually rely on single-input models, which can often result in the loss of fine anatomical details or important contextual information. In contrast, our Dual-Input U-Net is specifically designed to process both the original low-resolution image and the super-resolved image simultaneously. This dual-path setup allows the network to extract high-frequency textures from the SRCNN output while preserving the contextual integrity of the original image. As a result, our model delivers improved perceptual quality and preserves fine details across complex regions such as mucosal folds, shadows, and areas affected by motion blur. Below, we compare our hybrid approach with two recent state-of-the-art methods in the table 4.

Table 4. CONTRASTS OUR DESIGN AGAINST RECENT STATE-OF-THE-ART MODELS

Paper Name	Architecture Overview	PSNR/SSIM	Remarks
This Paper	SRCNN + Dual-Input U-Net (two-stage hybrid pipeline)	38.77 / 0.98	Combines resolution enhancement and structural refinement; flexible & extendable.
A Lightweight CNN for Detail Enhancement and Color Correction of Low-Light Capsule Endoscopy [21]	Lightweight CNN with IGEB and TCRB blocks	39.17 / 0.9792	Compact and fast; focused on low-light WCE images but less extensible or interpretable.
Global and Local Enhancement of Low-light Endoscopic Images [16]	Curve-based global module + dual attention + denoising	27.32 / 0.8349	End-to-end model designed for uneven illumination correction in low-light endoscopy.

While our approach slightly trails method proposed by Shuocheng Wang et al [21] in PSNR, it surpasses in structural similarity (SSIM) and offers impressive flexibility

thanks to its modular design. Besides, our pipeline allows for customized post-processing steps that improve clinical interpretability, something end-to-end models often struggle to provide. This hybrid strategy provides a good middle ground, balancing clarity, resolution, and adaptability, making it especially suitable for practical medical imaging applications.

4 Results

The evaluation of the deep learning based enhancement framework unfolds in this section. The assessment of the model focuses on quantitative evaluation alongside qualitative visual analysis before performing an extensive analysis of positive points, drawbacks, and possible enhancements.

4.1 Evaluation Metrics

The performance evaluation of the enhancing technique relies on two established image quality assessment metrics. Image quality assessment begins with the PSNR, which measures the relation between signal power capacity and noise-related power that degrades image quality. Higher PSNR values mean the image quality improves because there's less distortion present.

Mathematically, PSNR is defined as:

$$PSNR = 10\log_{10}\left(\frac{MAX_I^2}{MSE}\right) \qquad (2)$$

where $\boldsymbol{MAX_I}$ The maximum possible pixel value of the image (e.g., 255 for 8-bit images), and MSE denotes the Mean Squared Error between the original and the enhanced images [24, 27].

The SSIM is a commonly used metric for assessing image quality. It compares two images based on how similar they look to the human eye, focusing on aspects like brightness, contrast, and the overall structure of the details. In contrast to PSNR, which focuses on individual pixels, SSIM offers a more comprehensive evaluation of image fidelity by integrating principles of human visual perception.

SSIM is mathematically defined as:

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \qquad (3)$$

where $\mu x, \mu y$ Represent mean intensities, σx, σy Denote variances, and σxy Indicates the covariance. $C1$ and $C2$ These are small constants that stabilize the division to avoid instabilities when denominators are close to zero [24].

The SRCNN model demonstrates substantial improvement in image resolution, achieving a PSNR of 38.77 dB and an SSIM of 0.98, indicating high fidelity to the

original input and excellent structural preservation. Subsequently, the Dual-Input U-Net contributes further refinement by enhancing fine details and texture continuity. Although this stage results in a lower PSNR of 31.18 dB and an SSIM of 0.89, the perceptual quality and anatomical integrity of the images are visibly improved. Following the final post-processing stage, which includes contrast enhancement, sharpening, and noise reduction, the metrics decrease to a PSNR of 23.97 dB and SSIM of 0.77. The decrease is mainly due to bold pixel changes, but overall, the image is easier to understand and better for diagnosis. This shows that the method is useful in real clinical situations.

4.2 Visual Comparison

In addition to the numerical metrics, perceptual image quality plays a critical role in medical diagnosis, as it directly influences clinicians' ability to interpret endoscopic images. To assess the effectiveness of the proposed enhancement methods, a visual comparison is made between the original endoscopic images and those enhanced by the SRCNN and Dual-Input U-Net. This comparison demonstrates clear improvements in image sharpness, contrast, and the smoothness of textures, as illustrated in Figure 6. These visual evaluations are essential for gauging how the improvements actually influence diagnostic processes in real-world settings.

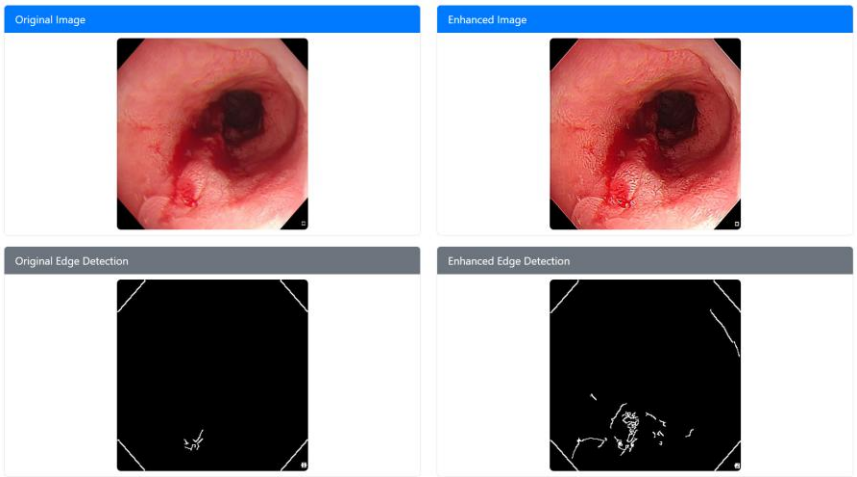

Fig. 6. Comparative visualization of endoscopic images before and after enhancement using SRCNN and Dual-Input U-Net.

Observations from Visual Comparisons:

- Raw Endoscopic Images: The original input images exhibit noticeable blur, low contrast, and considerable noise, all of which hinder accurate clinical interpretation and reduce diagnostic reliability.

- Enhanced Image: The image maintains clarity in structural details, smooth transitions in texture, and strong contrast. Small anatomical features become more noticeable, thanks to the improved resolution from upscaling, which also brings sharper edge definition. However, residual noise and mild blurriness remain, especially in complex regions, which will be tackled by post-processing.

4.3 Effect of Post-Processing Techniques

In addition to deep learning-based enhancement, a series of post-processing techniques are applied to further refine image clarity. These methods help make the images clearer by improving contrast, fixing uneven lighting, and reducing noise. This makes the pictures easier to see clearly and better for medical diagnosis.

Table 4. EXAMPLES OF POST-PROCESSING IMPACT ON IMAGE QUALITY

Post-Processing Technique	PSNR (dB) ↑	SSIM ↑	Effect on Image
Sharpening	+0.8	+0.03	Enhances fine edges and details
CLAHE (Illumination Correction)	+1.2	+0.04	Corrects uneven lighting
Gamma Correction	+0.7	+0.02	Adjusts brightness non-linearly

Fig. 7. Comparative analysis of different post-processing techniques on endoscopic images.

Key Findings from selected Post-Processing Analysis:

- CLAHE and White Balancing: These methods greatly enhance both the brightness and color accuracy of images, resulting in visuals that look more natural and provide better clarity for diagnostic purposes.
- Denoising Techniques: Noise is reduced so that the important parts of the anatomy stay clear, making sure no critical details get washed out when the image is improved.
- Bilateral Filtering: This approach enhances image smoothness and texture while maintaining sharp edge definition, balancing clarity and realism.

5 Discussion

Advantages of the Proposed Approach.

- Combining **SRCNN with Dual-Input U-Net** greatly enhances image resolution, reduces noise, and preserves structural details, surpassing traditional methods.
- **Edge-preserving enhancement** uses a special loss function that helps keep important boundaries in medical images clear. This way, the details don't get blurry or lost.
- **Highly resistant artifacts:** The technique demonstrates strong robustness when it comes to common challenges in endoscopy, such as motion blur, reflections from shiny surfaces, and varying lighting conditions.
- **Clinical Utility:** Enhancing the clarity of textures and reducing image noise allows for more precise detection of polyps, better segmentation of lesions, and increased overall reliability in diagnosing during gastrointestinal examinations.

Table 6. PROPOSED ARCITECTURE'S COMPUTATIONAL PROFILE

Component	Parameters (M)	Memory (approx.)	Avg Inference Time (ms)
SRCNN	~ 0.58	~ 120 MB	~ 8
Dual-Input U-Net	~ 3.5	~ 480 MB	~ 19
Post-Processing Stack	-	~ 50 MB	~ 12
Total	-	~ 650 MB	~ 39

These findings confirm that the model is both lightweight and fast, and is suitable for integration into clinical-grade image enhancement systems.

Limitations

- Dataset-Specific Training: The model has been primarily fine-tuned using specialized endoscopic datasets. This focus may restrict its applicability to other types of medical imaging or varied clinical environments.
- Over-Smoothing Risk: Intensive post-processing methods might unintentionally result in the loss of delicate textural details, which could affect the precise evaluation of pathological characteristics.

6 Conclusion

This study focuses on developing a deep learning-based image enhancement framework specifically designed to improve the quality of endoscopic images. The goal was to tackle common issues like low resolution, excessive noise, and inadequate contrast, making these images clearer and more useful for medical analysis. The proposed dual-stage setup, combining an SRCNN with a Dual-Input U-Net, noticeably enhances image sharpness, contrast, and the preservation of structural details [9, 16]. Quantitative assessments using PSNR and SSIM indicate notable enhancements in image quality [24, 27], confirming the model's capability to produce diagnostically valuable results. Applying post-processing techniques like sharpening, gamma correction, adjusting illumination, denoising, and white balancing can considerably enhance image clarity. These steps make it easier to interpret visuals accurately, especially in clinical settings where precision matters.

7 Future Scope

Improving Generalization Across Datasets

- Diverse Dataset Integration: Gathering information from multiple medical facilities helps cover a wider range of imaging scenarios and clinical variations.
- Domain Adaptation Techniques: Investigating techniques that can connect images captured from various endoscope manufacturers and environments, thus improving the model's resilience.
- Self-Supervised and Contrastive Learning: Using these approaches helps improve the model's ability to generalize without relying extensively on large labelled datasets, allowing it to pick up on fundamental features directly from unlabelled images.

Integration with AI-Assisted Diagnosis

- Automated Detection: This process focuses on creating object detection algorithms that can automatically identify artifacts, which in turn supports earlier and more precise diagnoses.

- Segmentation for Precise Localization: Using deep learning-based segmentation networks to accurately identify areas of interest, enabling more targeted and efficient clinical assessments.
- AI-Driven Decision Support Systems: Creating real-time decision support systems driven by AI, which use advanced imaging and diagnostic algorithms to provide clinicians with actionable insights throughout endoscopic procedures.